Programming Fundamentals

Lab 4 Assignment

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Class: BSCS

Section: A

Lab Task 1

Handwritten code

Task 1:

Source Code (cpp):

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;
    cout << "Enter a number: ";
    cin >> number;

    if (number >= 0)
    {
        cout << "Number entered is positive: " << number << endl;
    }
    else
    {
        cout << "Number is negative" << endl;
    }
    return 0;
}
```

Output:

```
Enter a number: 7
Number entered is positive: 7

Enter a number: -4
Number is negative
```

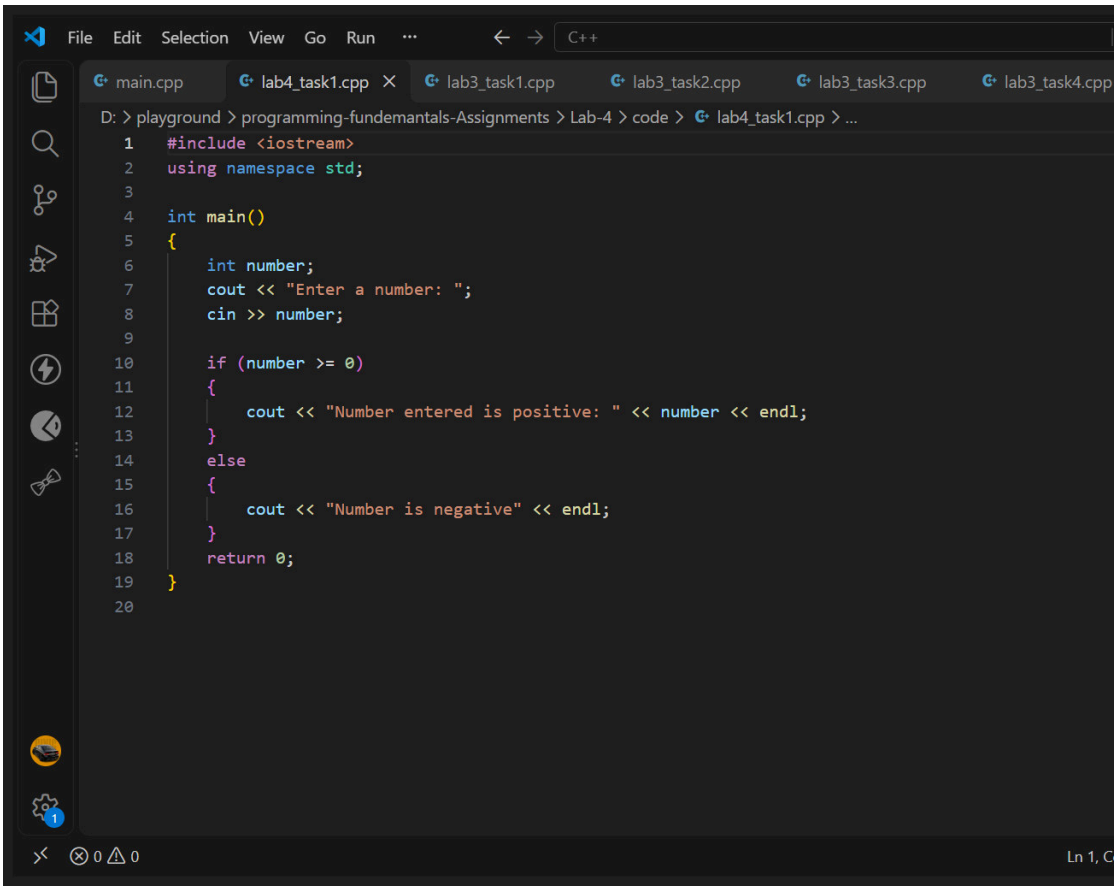
Output (example):

```
Enter a number: 7
Number entered is positive: 7

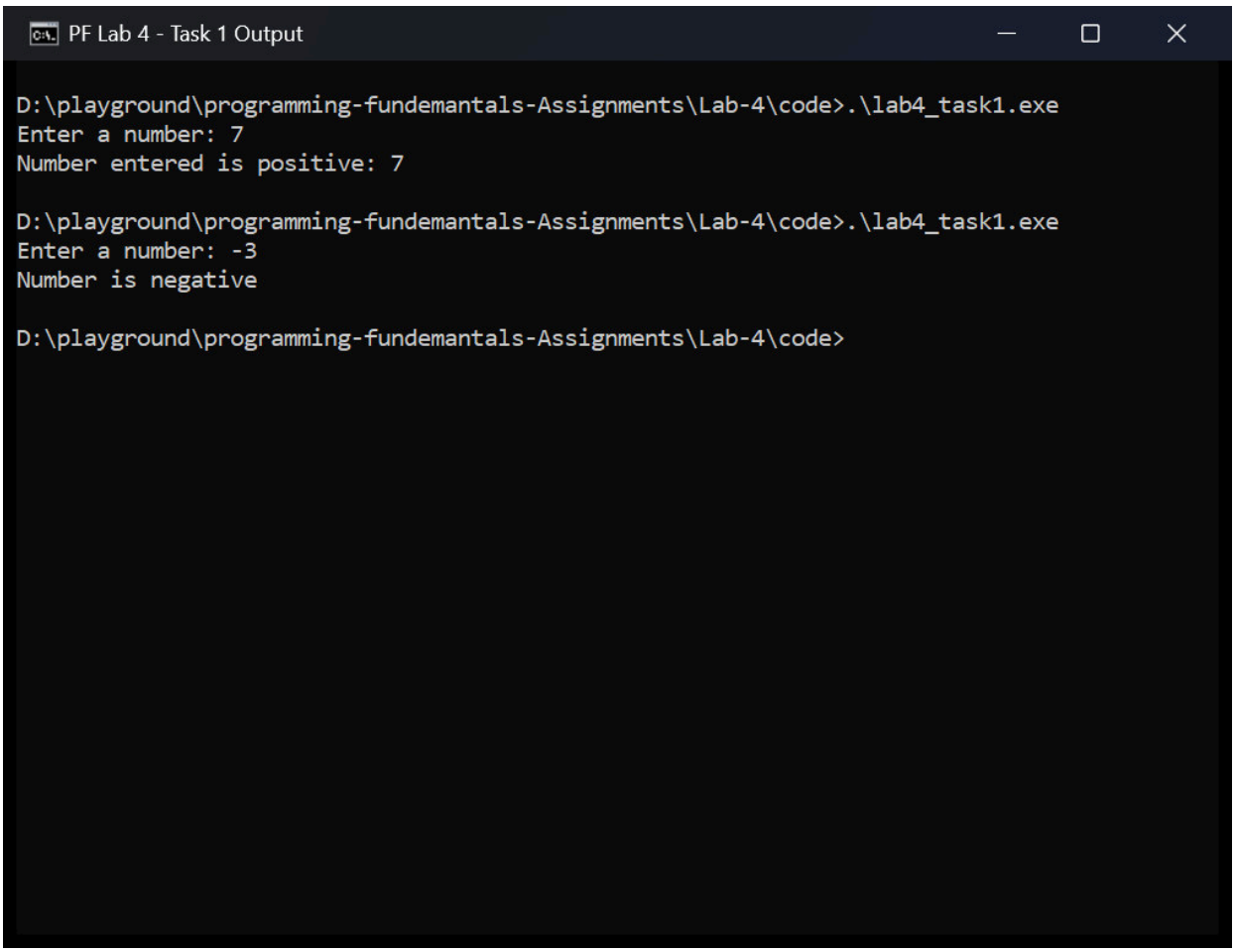
Enter a number: -4
Number is negative
```

Lab Task 1

Typed code (lab4_task1.cpp)

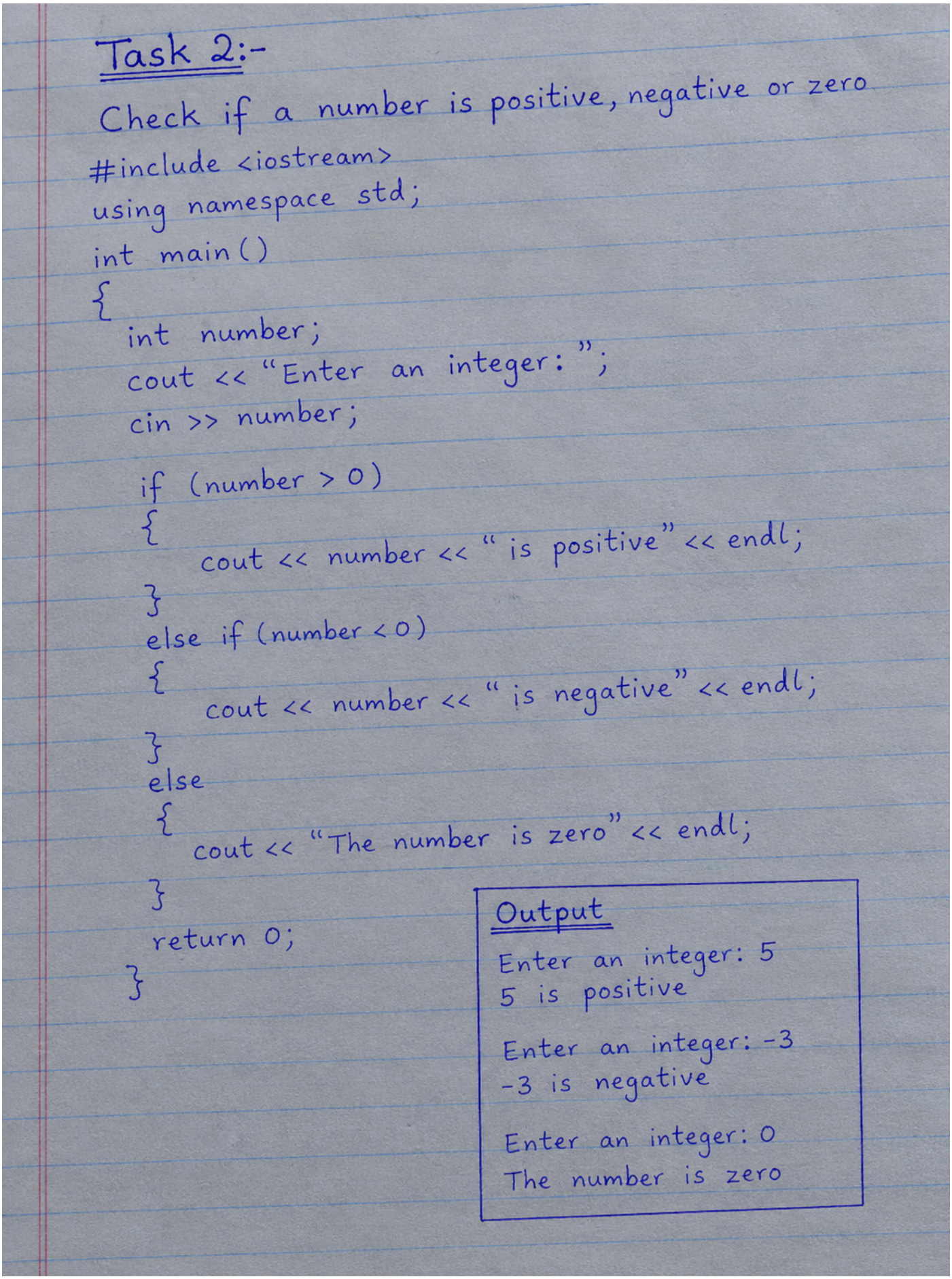


Output



Lab Task 2

Handwritten code



Task 2:-

Check if a number is positive, negative or zero

```
#include <iostream>
using namespace std;
int main()
{
  int number;
  cout << "Enter an integer: ";
  cin >> number;

  if (number > 0)
  {
     cout << number << " is positive" << endl;
  }
  else if (number < 0)
  {
     cout << number << " is negative" << endl;
  }
  else
  {
     cout << "The number is zero" << endl;
  }
  return 0;
}
```

Output

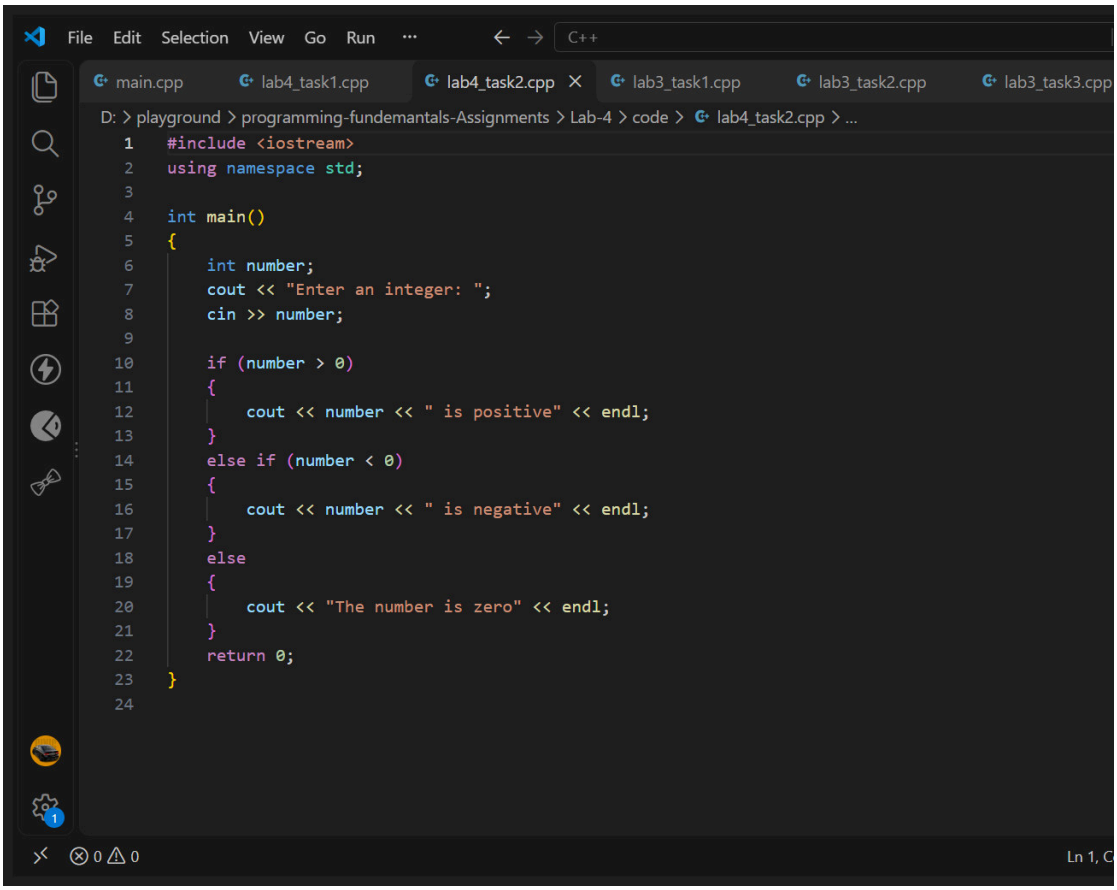
```
Enter an integer: 5
5 is positive

Enter an integer: -3
-3 is negative

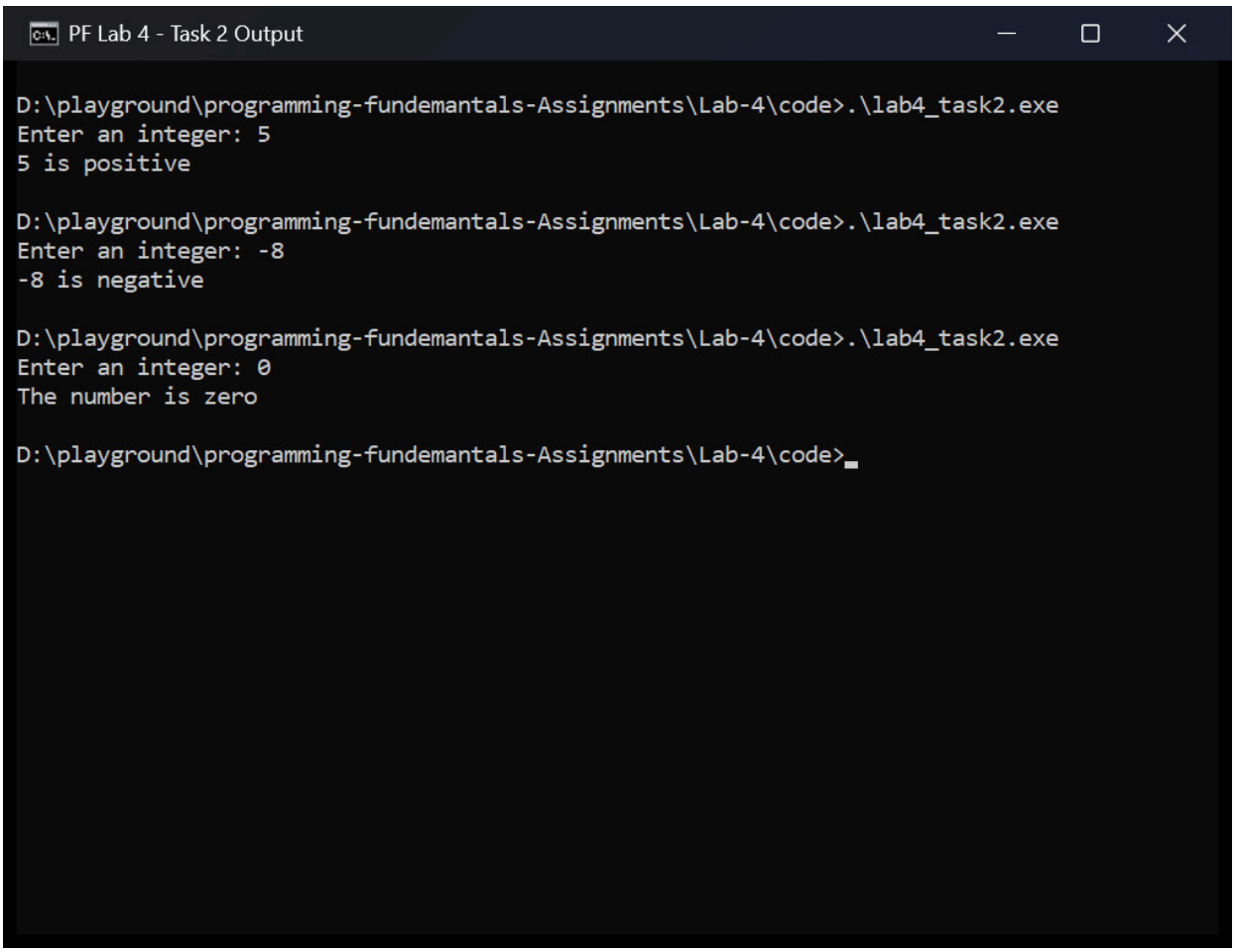
Enter an integer: 0
The number is zero
```

Lab Task 2

Typed code (lab4_task2.cpp)

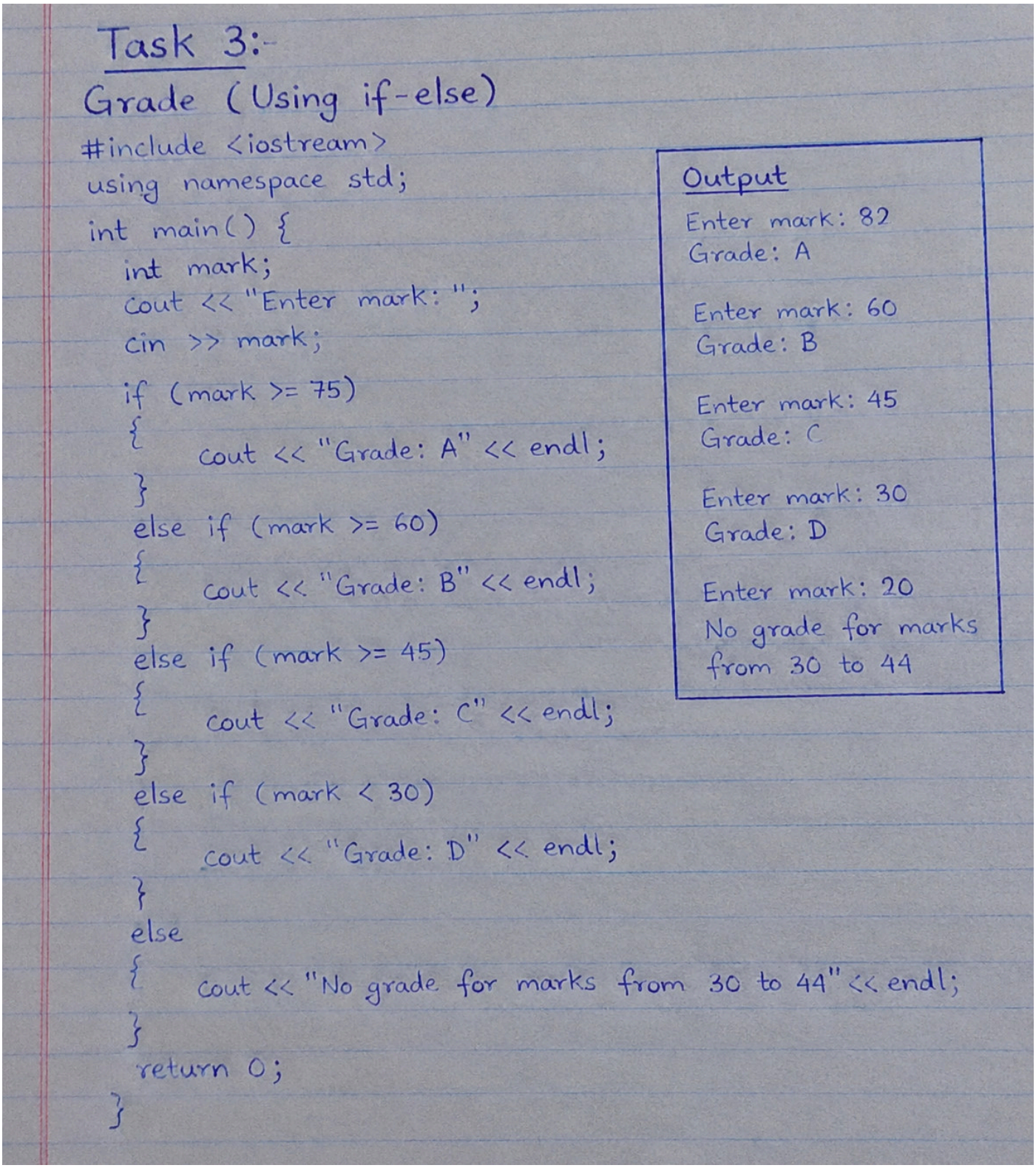


Output



Lab Task 3

Handwritten code



Task 3:-

Grade (Using if-else)

```
#include <iostream>
using namespace std;
int main() {
    int mark;
    cout << "Enter mark: ";
    cin >> mark;
    if (mark >= 75)
    {
        cout << "Grade: A" << endl;
    }
    else if (mark >= 60)
    {
        cout << "Grade: B" << endl;
    }
    else if (mark >= 45)
    {
        cout << "Grade: C" << endl;
    }
    else if (mark < 30)
    {
        cout << "Grade: D" << endl;
    }
    else
    {
        cout << "No grade for marks from 30 to 44" << endl;
    }
    return 0;
}
```

Output

```
Enter mark: 82
Grade: A

Enter mark: 60
Grade: B

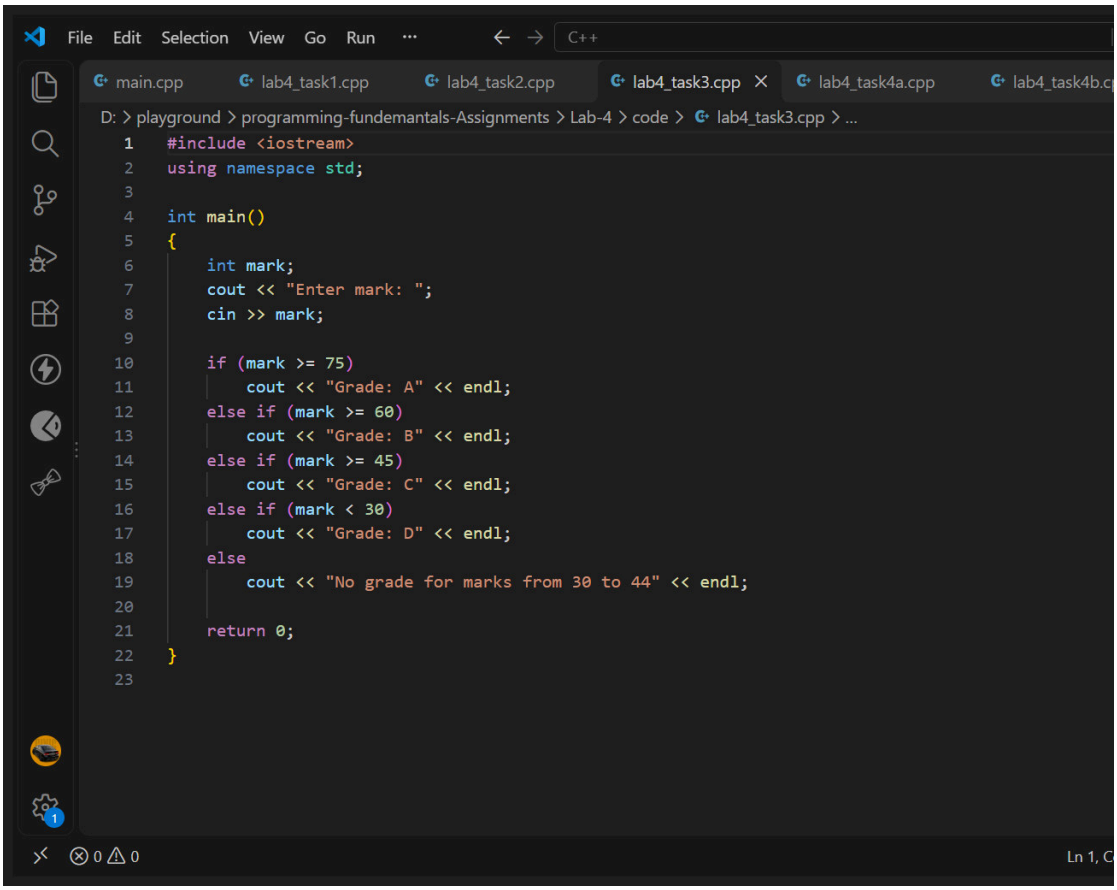
Enter mark: 45
Grade: C

Enter mark: 30
Grade: D

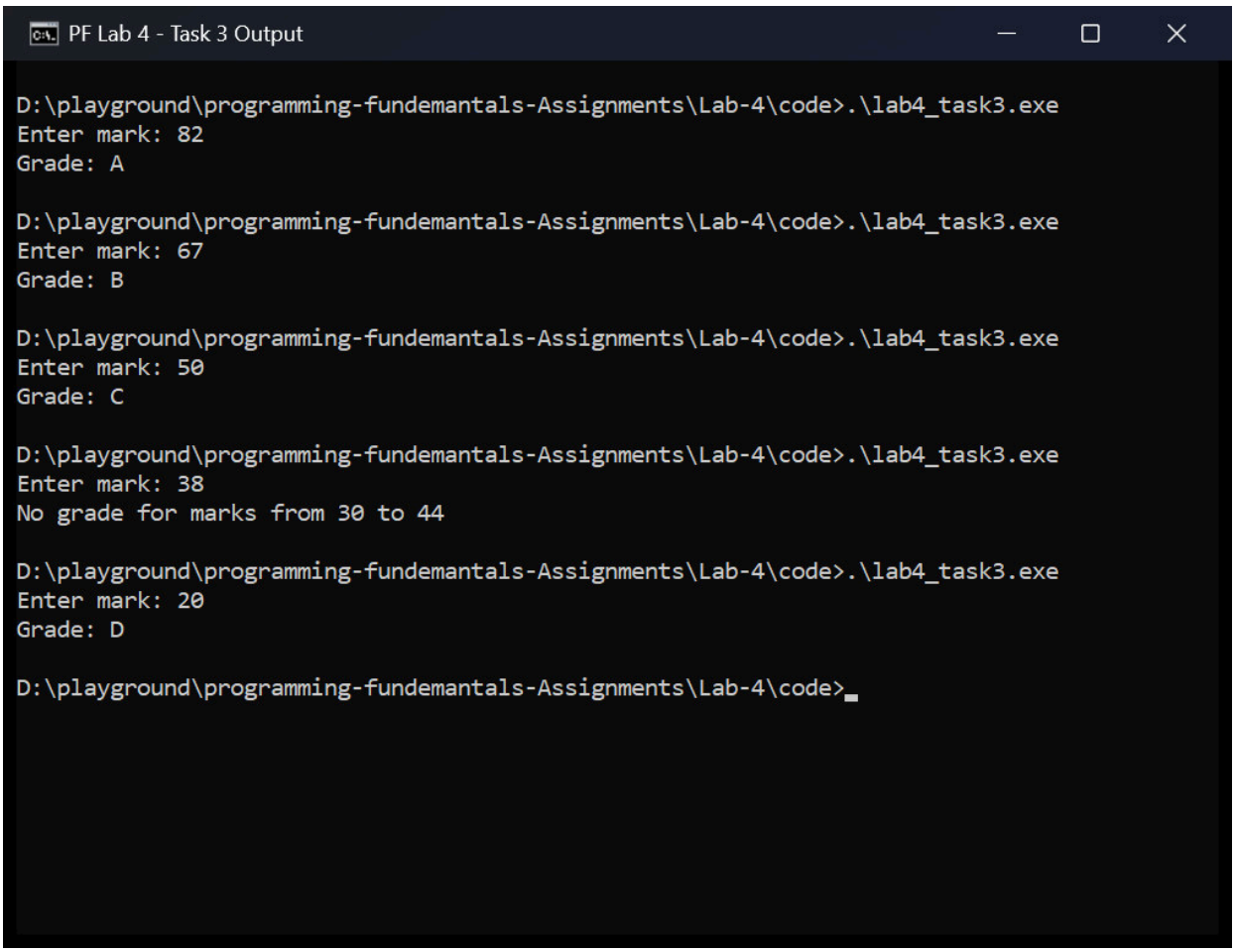
Enter mark: 20
No grade for marks
from 30 to 44
```

Lab Task 3

Typed code (lab4_task3.cpp)

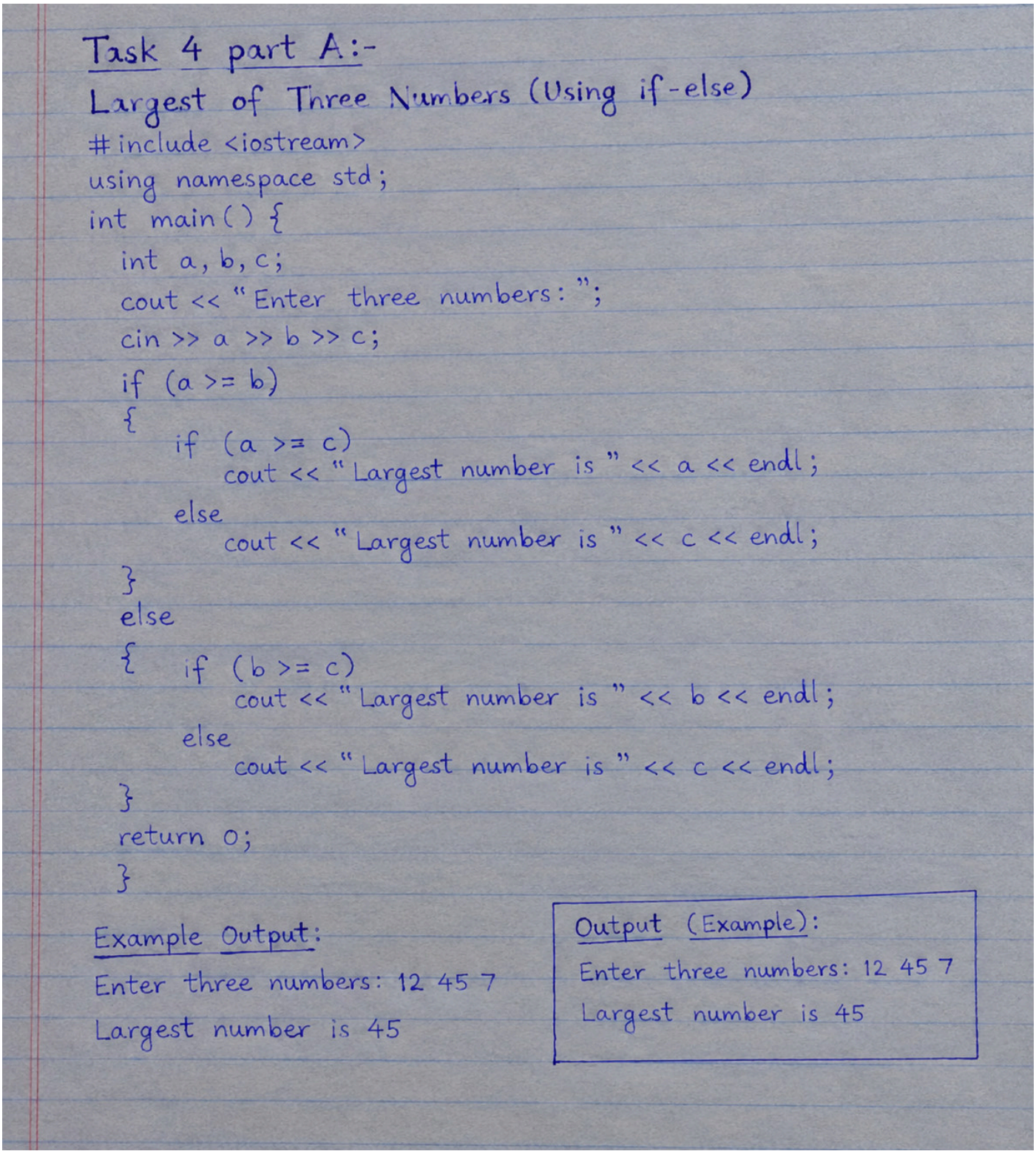


Output



Lab Task 4 - Part A

Handwritten code



Task 4 part A:-

Largest of Three Numbers (Using if-else)

```
#include <iostream>
using namespace std;
int main() {
  int a, b, c;
  cout << "Enter three numbers: ";
  cin >> a >> b >> c;
  if (a >= b)
  {
      if (a >= c)
          cout << "Largest number is " << a << endl;
      else
          cout << "Largest number is " << c << endl;
  }
  else
  {   if (b >= c)
          cout << "Largest number is " << b << endl;
      else
          cout << "Largest number is " << c << endl;
  }
  return 0;
  }
```

Example Output:

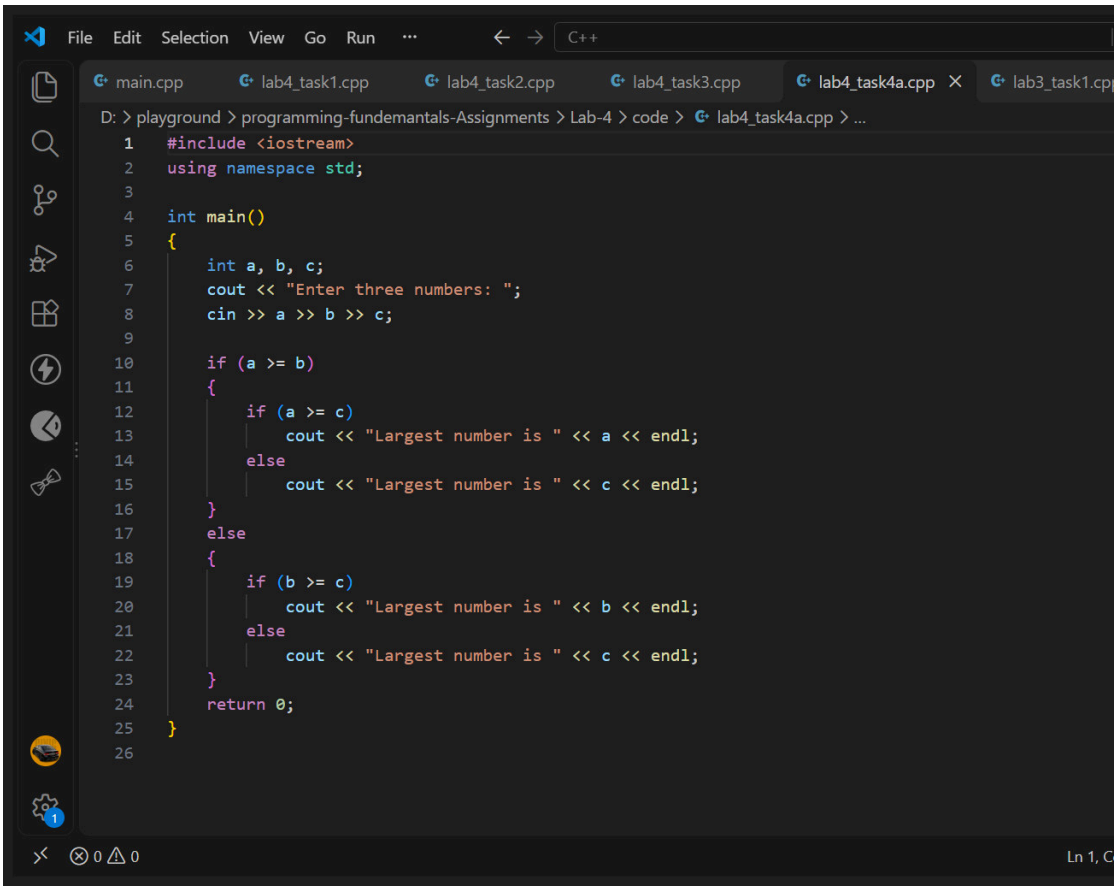
```
Enter three numbers: 12 45 7
Largest number is 45
```

Output (Example):

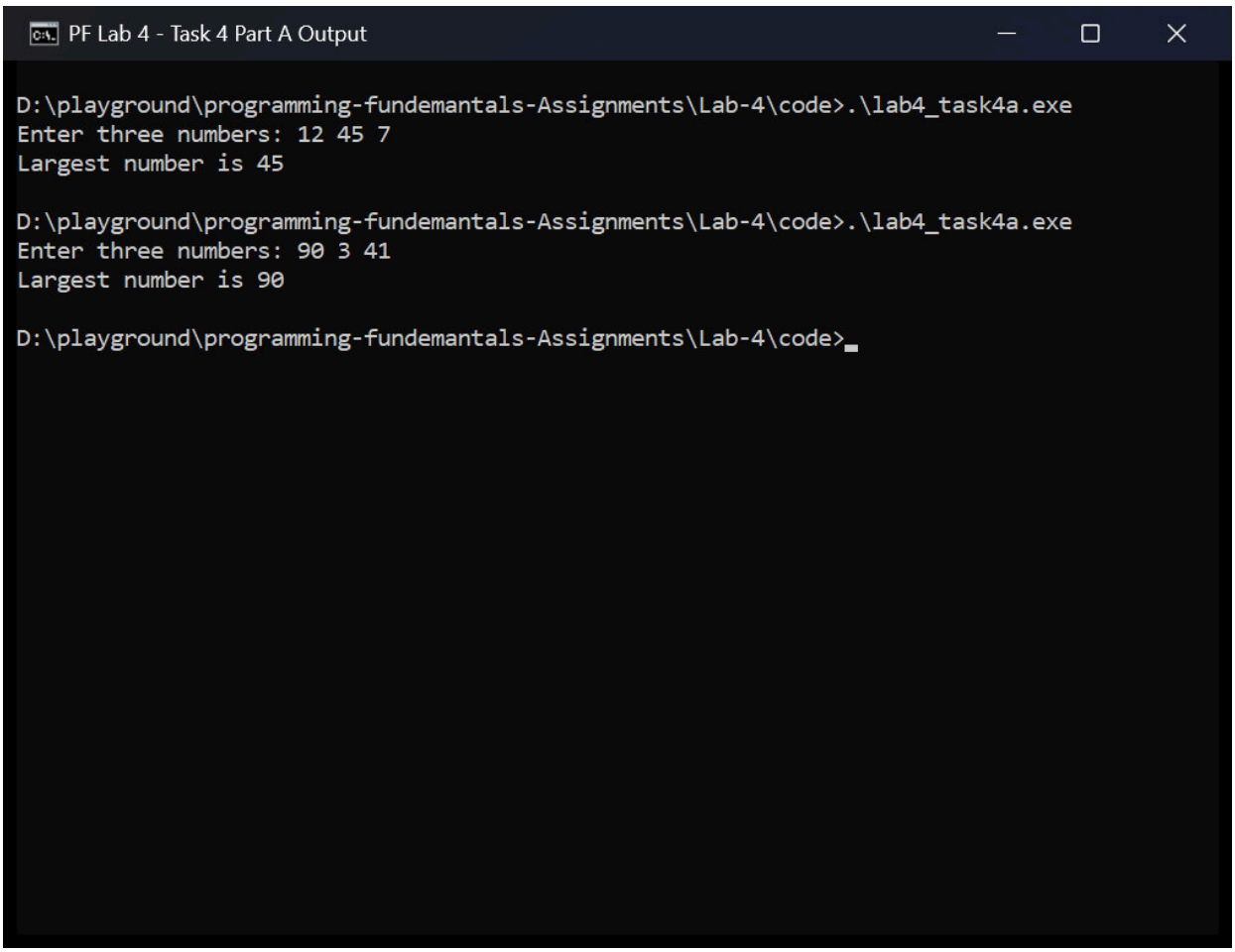
```
Enter three numbers: 12 45 7
Largest number is 45
```

Lab Task 4 - Part A

Typed code (lab4_task4a.cpp)



Output



Lab Task 4 - Part B

Handwritten code

Task 4 part B:-

Check if a number has 4 digits (Using if-else)

```
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number > 0) {
        if (number < 10)
            cout << "It has 1 digit" << endl;
        else if (number < 100)
            cout << "It has 2 digits" << endl;
        else if (number < 1000)
            cout << "It has 3 digits" << endl;
        else if (number < 10000)
            cout << "It has 4 digits" << endl;
        else
            cout << "It has 5 or more digits" << endl;
    }
    else
        cout << "It has 5 or more digits" << endl;
    }
    return 0;
}
```

Example Output:

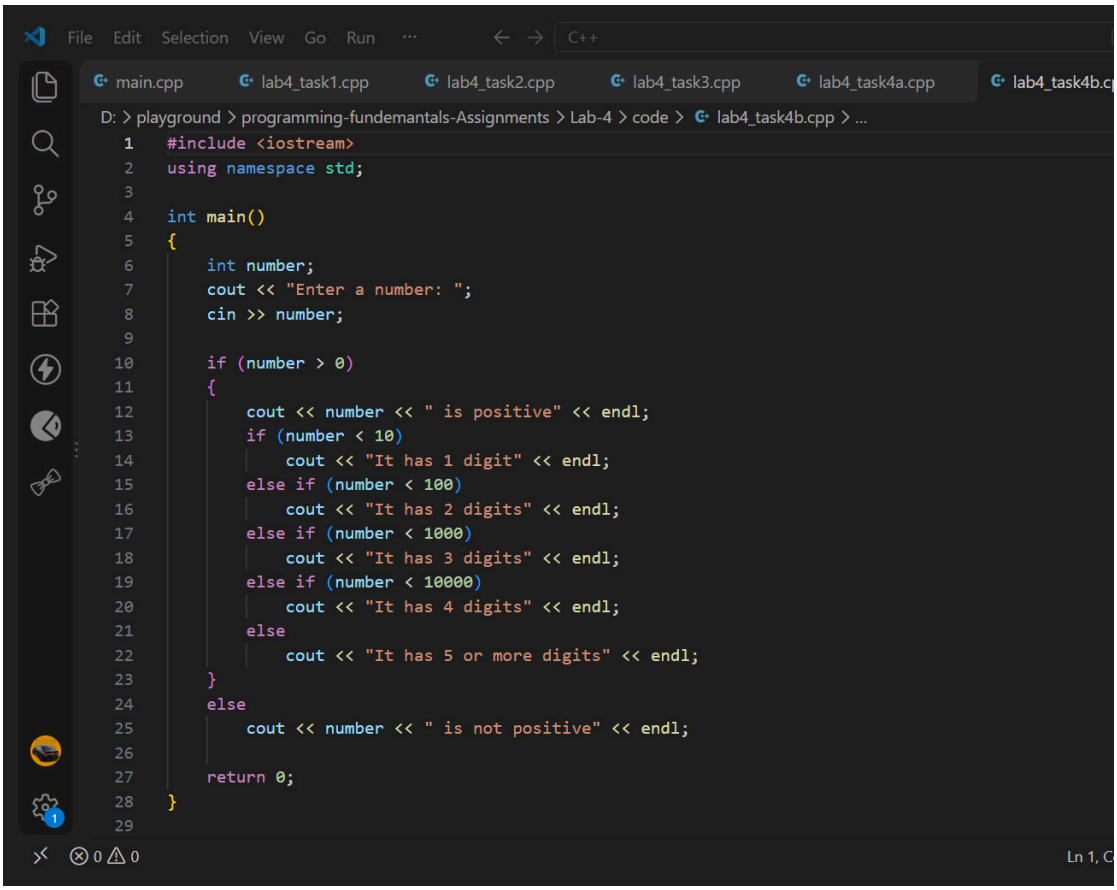
```
Enter a number: 4567
4567 is positive
It has 4 digits
```

Output (Example):

```
Enter a number: 4567
4567 is positive
It has 4 digits
```

Lab Task 4 - Part B

Typed code (lab4_task4b.cpp)



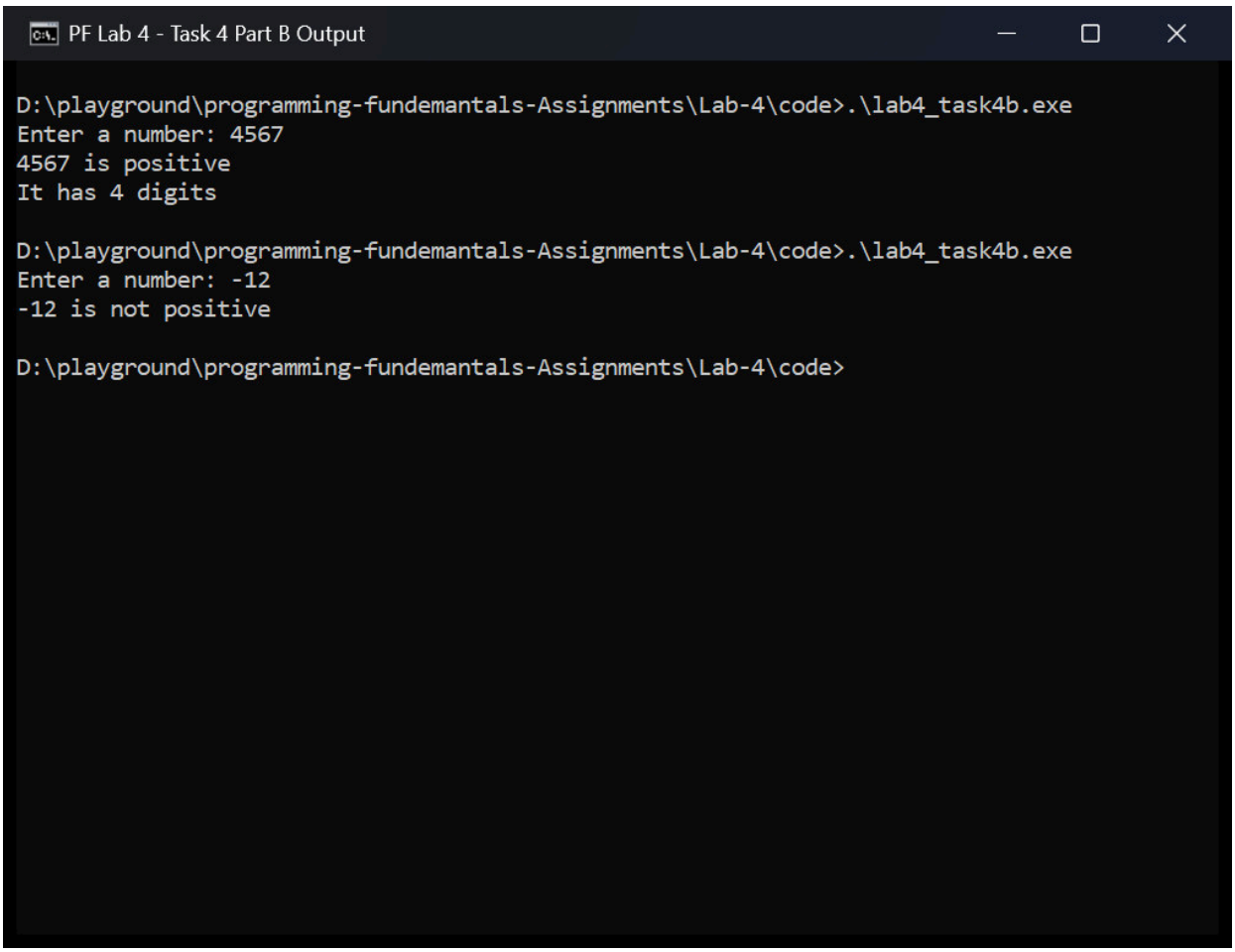
```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;
    cout << "Enter a number: ";
    cin >> number;

    if (number > 0)
    {
        cout << number << " is positive" << endl;
        if (number < 10)
            cout << "It has 1 digit" << endl;
        else if (number < 100)
            cout << "It has 2 digits" << endl;
        else if (number < 1000)
            cout << "It has 3 digits" << endl;
        else if (number < 10000)
            cout << "It has 4 digits" << endl;
        else
            cout << "It has 5 or more digits" << endl;
    }
    else
        cout << number << " is not positive" << endl;

    return 0;
}
```

Output



```
PF Lab 4 - Task 4 Part B Output

D:\playground\programming-fundemantals-Assignments\Lab-4\code>.\lab4_task4b.exe
Enter a number: 4567
4567 is positive
It has 4 digits

D:\playground\programming-fundemantals-Assignments\Lab-4\code>.\lab4_task4b.exe
Enter a number: -12
-12 is not positive

D:\playground\programming-fundemantals-Assignments\Lab-4\code>
```